EE3110 - Probability Foundations for Electrical Engineers
Tutorial - 3

1. Let X be a random day of the week, coded so that Monday is 1, Tuesday is 2, etc. (so X takes values 1, 2,..., 7, with equal probabilities). Let Y be the next day after X (again represented as an integer between 1 and 7). Do X and Y have the same distribution? What is $P(X < Y)$?

Solution: They have the same distribution since Y is also equally likely to represent any day of the week, but $P(X < Y) = P(X \neq 7) = 6/7$.

2. Let X and Y be two independent $Ber(\frac{1}{2})$ random variables. Define random variables U and V by:

$$U = X + Y \text{ and } V = |X - Y|$$

   1. Determine the joint and marginal probability distributions of U and V.
   2. Find out whether U and V are dependent or independent.

Solution:

1. The joint PDF looks like this:

$V \backslash U$	0	1	2
0	0.25	0	0.25
1	0	0.5	0

   The marginal probability distribution for U is given by:

$$P_U(0) = \frac{1}{4}, P_U(1) = \frac{1}{2}, P_U(2) = \frac{1}{4}$$

   and for V as follows:

$$P_V(0) = P_V(1) = \frac{1}{2}$$

2. Consider $U = 0$, $V = 1$. Then $P_{U,V}(1, 0) \neq P_U(1) \cdot P_V(0)$. So, U and V are dependent.

3. Let X be a random variable with law $f(x) = C\frac{2^x}{x!}$ for $x \in \mathbb{N}$. Then, the probability that X is even is approximately _____.

Solution: We have,

$$\mathbb{P}[\Omega] = 1$$
$$\Rightarrow \sum_{x=1}^{\infty} f(x) = \sum_{x=1}^{\infty} C\frac{2^x}{x!} = 1$$

Solving for C, we obtain $C = \frac{1}{e^2-1}$
To find $\mathbb{P}[X \textit{ is even}] = \mathbb{P}[X = 2k : k \in \mathbb{N}]$

Therefore,

$$\mathbb{P}\left[X \text{ is even}\right] = \sum_{k=1}^{\infty} C\frac{2^{2k}}{(2k)!} = C\sum_{i=0}^{\infty}\frac{2^i + (-2)^i}{2(i!)} - 1$$
$$= C\left(\frac{e^2 + e^{-2}}{2} - 1\right)$$
$$= \frac{1 - e^{-2}}{2}$$

4. **Happy Birthday again!** You go to a party with 500 guests. What is the probability that exactly one other guest has the same birthday as you? Calculate this exactly and also approximately by using the Poisson PMF. (For simplicity, exclude birthdays on February 29.)

Solution: The number of guests that have the same birthday as you is binomial with p= 1/365 and n= 499. Thus, the probability that exactly one other guest has the same birthday is

$$\binom{499}{1} \times \frac{1}{365} \times \left(\frac{364}{365}\right)^{498}$$

Let $\lambda = np = 499/365 \approx 1.367$. The Poisson approximation is $e^{-\lambda}\lambda = e^{-1.367} \times 1.367 \approx 0.3483$, which closely agrees with the correct probability based on the binomial.

5. **Memorylessness** Let X be a discrete random variable following the Geometric PMF with parameter p. Show the following:

$$\mathbb{P}(X = n + k | X > n) = \mathbb{P}(X = k). \tag{1}$$

This means that given you have a already waited for n trails for a success the probability that you will wait for k more trails is same as the probability of waiting for k trails from the first trail, i.e., the random variable doesn't remember the past.

Solution: For X which is geometric with parameter p, its PMF is :

$$\mathbb{P}(X = k) = (1-p)^{k-1}p \quad k = 1, 2, 3.....$$

$$\mathbb{P}(X = n + k | X > n) = \frac{\mathbb{P}(X = n + k, X > n)}{\mathbb{P}(X > n)}$$

The intersection of the events $X = n + k$ and $X > n$, is simply $X = n + k$

$$\mathbb{P}(X = n + k | X > n) = \frac{\mathbb{P}(X = n + k)}{\mathbb{P}(X > n)}$$

$$\mathbb{P}(X > n) = \sum_{i=n+1}^{\infty} \mathbb{P}(X = i) = \sum_{i=n+1}^{\infty} (1-p)^{i-1}p$$

Let $j = i - n$, then

$$\mathbb{P}(X > n) = \sum_{j=1}^{\infty}(1-p)^{j+n-1}p = (1-p)^n \sum_{j=1}^{\infty}(1-p)^{j-1}p = (1-p)^n \cdot 1$$

$$\begin{aligned}\mathbb{P}(X = n+k | X > n) &= \frac{\mathbb{P}(X = n+k)}{\mathbb{P}(X > n)} \\ &= \frac{(1-p)^{n+k-1}p}{(1-p)^n} \\ \therefore \mathbb{P}(X = n+k | X > n) &= (1-p)^{k-1}p = \mathbb{P}(X = K)\end{aligned}$$

6. The joint probability distribution of two discrete random variables X and Y is given by:

$X \backslash Y$	y_1	y_2	y_3
x_1	0.10	0.20	0.10
x_2	0.15	0.25	0.20

1. **Marginal distributions**
   (a) Find the marginal distribution of X.
   (b) Find the marginal distribution of Y.
2. **Conditional distributions**
   (a) Compute $P(X = x_1 \mid Y = y_2)$.
   (b) Compute $P(Y = y_3 \mid X = x_2)$.
3. **Theorem of Total Probability (verification)**

   Verify that

   $$P(X = x_i) = \sum_j P(X = x_i \mid Y = y_j)\, P(Y = y_j)$$

   by carrying out the calculation explicitly for $X = x_1$.

Solution:

1. Marginal distributions

(a) Marginal of X.

$$P(X = x_1) = 0.10 + 0.20 + 0.10 = 0.40,$$
$$P(X = x_2) = 0.15 + 0.25 + 0.20 = 0.60.$$

(b) Marginal of Y.

$$P(Y = y_1) = 0.10 + 0.15 = 0.25,$$
$$P(Y = y_2) = 0.20 + 0.25 = 0.45,$$
$$P(Y = y_3) = 0.10 + 0.20 = 0.30.$$

2. Conditional distributions

(a) $P(X = x_1 \mid Y = y_2)$:

$$P(X = x_1 \mid Y = y_2) \;=\; \frac{P(X = x_1, Y = y_2)}{P(Y = y_2)} \;=\; \frac{0.20}{0.45} \;=\; \frac{4}{9} \;\approx\; 0.4444.$$

(b) $P(Y = y_3 \mid X = x_2)$:

$$P(Y = y_3 \mid X = x_2) \;=\; \frac{P(X = x_2, Y = y_3)}{P(X = x_2)} \;=\; \frac{0.20}{0.60} \;=\; \frac{1}{3} \;\approx\; 0.3333.$$

3. Theorem of Total Probability verification

$$P(X = x_1 \mid Y = y_1) = \frac{0.10}{0.25} = 0.4, \qquad 0.4 \cdot 0.25 = 0.10,$$

$$P(X = x_1 \mid Y = y_2) = \frac{0.20}{0.45} = \frac{4}{9} \approx 0.4444, \qquad \frac{4}{9} \cdot 0.45 = 0.20,$$

$$P(X = x_1 \mid Y = y_3) = \frac{0.10}{0.30} = \frac{1}{3} \approx 0.3333, \qquad \frac{1}{3} \cdot 0.30 = 0.10.$$

Summing:

$$0.10 + 0.20 + 0.10 = 0.40 = P(X = x_1),$$

which verifies the theorem for $X = x_1$.

7. Alice flips a fair coin n times and Bob flips another fair coin $n + 1$ times, resulting in independent $X \sim \text{Bin}(n, \frac{1}{2})$ and $Y \sim \text{Bin}(n + 1, \frac{1}{2})$.

   (a) Show that $P(X < Y) = P(n - X < n + 1 - Y)$.

   (b) Compute $P(X < Y)$.

Solution: (a) Note that $n - X \sim \text{Bin}(n, 1/2)$ and $n + 1 - Y \sim \text{Bin}(n + 1, 1/2)$ (we can interpret this by thinking of counting Tails rather than counting Heads), with $n - X$ and $n + 1 - Y$ independent. So

$$P(X < Y) = P(n - X < n + 1 - Y),$$

since both sides have exactly the same structure.

(b) We have

$$P(X < Y) = P(n - X < n + 1 - Y) = P(Y < X + 1) = P(Y \leq X)$$

since X and Y are integer-valued (e.g., $Y < 5$ is equivalent to $Y \leq 4$). But $Y \leq X$ is the complement of $X < Y$, so

$$P(X < Y) = 1 - P(X < Y).$$

Thus,

$$P(X < Y) = \frac{1}{2}.$$

8. **Probability of Triangle** Two points are chosen with uniform probability and independently on a stick. It is broken at those two points to form three segments. What is the probability that the three segments can form a triangle?

Solution: Without loss of generality, assume that the length of the stick is 1. Let the point of the first cut be x and the second cut be y, with $0 < x < y < 1$.

The three segments are:

$$x, \quad y - x, \quad 1 - y.$$

For these to form a triangle, the triangle inequality must hold:

$$\begin{aligned} x + (y - x) > 1 - y \quad &\Rightarrow \quad y > \tfrac{1}{2}, \\ x + (1 - y) > y - x \quad &\Rightarrow \quad y < x + \tfrac{1}{2}, \\ (y - x) + (1 - y) > x \quad &\Rightarrow \quad x < \tfrac{1}{2}. \end{aligned}$$

Thus, the valid region for (x, y) is defined by:

$$0 < x < \tfrac{1}{2}, \quad \tfrac{1}{2} < y < x + \tfrac{1}{2}.$$

The area of the whole square $(0, 1) \times (0, 1)$ is 1. The feasible region is a right triangle of area $\frac{1}{8}$. By symmetry (since we could also have $y < x$), the total valid area is $\frac{1}{4}$.

Therefore, the required probability is

$$\frac{1}{4}.$$

9. Let $X_1, X_2, \ldots, X_n$ be i.i.d. real-valued random variables with common CDF $F(x) = \mathbb{P}(X_1 \leq x)$. Define the maximum $M = \max_{1 \leq i \leq n} X_i$ and the minimum $m = \min_{1 \leq i \leq n} X_i$. Derive the CDFs $F_M(x) = \mathbb{P}(M \leq x)$ and $F_m(x) = \mathbb{P}(m \leq x)$.

Solution:
Maximum.

$$\begin{aligned} F_M(x) = \mathbb{P}(M \leq x) &= \mathbb{P}\Big(\bigcap_{i=1}^{n} \{X_i \leq x\} \Big) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq x) \quad \text{(i.i.d.)} \\ &= \big(F(x)\big)^n. \end{aligned}$$

Minimum.

$$\begin{aligned} F_m(x) = \mathbb{P}(m \leq x) &= 1 - \mathbb{P}(m > x) = 1 - \mathbb{P}\Big(\bigcap_{i=1}^{n} \{X_i > x\} \Big) \\ &= 1 - \prod_{i=1}^{n} \mathbb{P}(X_i > x) = 1 - \big(1 - F(x)\big)^n. \end{aligned}$$

These identities hold for any CDF F (no continuity assumption needed).

10. A packet communication system consists of a buffer that stores packets from some source and a communication line that retrieves packets from the buffer and transmits them to a receiver. The communication system operates in time-slot pairs. In the first slot, the system stores some packets that are generated by the source according to a Poisson PMF with parameter λ. However, the maximum number of packets

that can be stored is b (some integer), and packets arriving after the buffer is full are discarded. In the second slot, the system transmits either all the stored packets or c ($0 < c < b$) packets (whichever is less).

1. Assuming that at the beginning of the first slot the buffer is empty, find the PMF of the number of packets stored at the end of the first slot and at the end of the second slot.
2. What is the probability that some packets get discarded during the first slot?

Solution:

(a) Let X be the number of packets stored at the end of the first slot. For $k < b$, the probability that X = k is the same as the probability that the source generates k packets and for $k = b$ it is the summation of probabilities of the source generating greater than or equal to k packets:

$$p_X(k) = e^{-\lambda}\frac{\lambda^k}{k!}, \quad k < b$$

$$p_X(b) = \sum_{k=b}^{\infty} e^{-\lambda}\frac{\lambda^k}{k!} = 1 - \sum_{k=0}^{b-1} e^{-\lambda}\frac{\lambda^k}{k!},$$

Let Y be the number of of packets stored at the end of the second slot. Since $min\{X, c\}$ is the number of packets transmitted in the second slot, we have

$$Y = X - min\{X, c\}$$

$$p_Y(0) = \sum_{k=0}^{c} p_X(k) = \sum_{k=0}^{c} e^{-\lambda}\frac{\lambda^k}{k!},$$

$$p_Y(k) = p_X(k+c) = e^{-\lambda}\frac{\lambda^{k+c}}{(k+c)!}, \quad 0 < k < b - c$$

$$p_Y(b-c) = p_X(b) = 1 - \sum_{k=0}^{b-1} e^{-\lambda}\frac{\lambda^k}{k!}$$

(b) The probability that some packets get discarded during the first slot is the same as the probability that more than b packets are generated by the source, so it is equal to

$$\sum_{k=b+1}^{\infty} e^{-\lambda}\frac{\lambda^k}{k!} = 1 - \sum_{k=0}^{b} e^{-\lambda}\frac{\lambda^k}{k!}$$

11. *(Buffon's Needle)* If a short needle of length l is dropped on a paper that is ruled with equally spaced lines of distance $d \geq l$, find the probability that the needle comes to lie in a position where it crosses one of the lines.
(Some trivia : This result was used as a measure of the value of pi experimentally in the 18th century.)

Solution: Let us say the needle drops to lie at an angle of θ away from the horizontal, where θ will be in the range of $0 \leq \theta \leq \pi/2$ (We will ignore the case where the needle comes to lie with

a negative slope, since that case is the same as that of positive slope and symmetric). The needle that lies with angle θ has a height $lsin\theta$, and the probability that such a needle crosses one of the horizontal lines of distance d is $\frac{lsin\theta}{d}$.

Thus, we get the probability by averaging over all the possible angles θ :

$$p = \frac{\int_0^{\frac{\pi}{2}} \frac{lsin\theta}{d} d\theta}{\int_0^{\frac{\pi}{2}} d\theta} = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \frac{lsin\theta}{d} d\theta = \frac{2l}{\pi d} [-cos\theta]_0^{\frac{\pi}{2}}$$

$$p = \frac{2l}{\pi d}$$

12. Arjun, Nishanth and Harshith are roommates. Every day they promise each other that they will sleep at 1:00 AM, but end up having life discussions, so they always sleep late.

- The time they **go to sleep** S can be modeled as a uniformly distributed **continuous** random variable between 1:00 AM and 6:00 AM, that is, $S \sim \text{Uniform}(1, 6)$.
- The time they **wake up** W can be modeled as a uniformly distributed **continuous** random variable between 8:00 AM and 12:00 PM, that is, $W \sim \text{Uniform}(8, 12)$.

Let us assume that S and W are independent.

(a) Every night, Arjun confidently proclaims that he will wake up at **exactly** 8:30 am and have breakfast. What is the probability that this happens?

(b) They have their first class at 9:00 AM. They will go to class if they wake up at or before 9:00 AM. What is the probability that they will go to class?

(c) They can stay awake in class if they get a minimum of 6 hours of sleep (basically just enough sleep). What is the probability that they will stay awake?

(d) DDiet is an eatery outside T Gate. If they are awake at 4:00 A.M. then they will go to DDiet to have early morning tea with a probability of 0.7. What is the probability that they go to DDiet and also have enough sleep?

Solution:

1. **Probability Arjun wakes at exactly 8:30 AM.**
The wake time has a distribution $W \sim \text{Uniform}(8, 12)$, so it has a probability density function

$$f_W(w) = \begin{cases} \frac{1}{4}, & w \in [8, 12], \\ 0, & \text{otherwise.} \end{cases}$$

If a random variable has a density, the probability of any single point a is the integral of the density over the degenerate interval $[a, a]$, hence

$$P(W = 8.5) = \int_{8.5}^{8.5} f_W(w)\, dw = 0.$$

Therefore $\boxed{P(W = 8.5) = 0}$.

2. **Probability of making it to class.** The wake time is $W \sim \text{Uniform}(8, 12)$. Hence

$$P(W \leq 9) = \frac{9-8}{12-8} = \frac{1}{4} = 0.25.$$

3. **Probability of getting at least 6 hours of sleep.** Now $S \sim \text{Uniform}(1, 6)$ and $W \sim \text{Uniform}(8, 12)$, independent. The joint sample region is the rectangle $(s, w) \in [1, 6] \times [8, 12]$ of area

$$(6-1) \cdot (12-8) = 5 \cdot 4 = 20.$$

For a fixed $w \in [8, 12]$, the condition $W - S \geq 6$ is $s \leq w - 6$. Since $w - 6 \in [2, 6]$ and $S \in [1, 6]$, the allowable s range for each w is $[1,\ w-6]$, which has length $(w-6)-1 = w-7$. Thus the favorable area is

$$\int_8^{12} (w-7)\, dw = \left[\tfrac{1}{2}w^2 - 7w\right]_8^{12} = 12.$$

Therefore

$$P(W - S \geq 6) = \frac{12}{20} = \frac{3}{5} = 0.6.$$

4. **Probability of going to DDiet and still getting enough sleep.** They go to DDiet with probability 0.7 *given* they are awake at 4:00 AM. Being awake at 4:00 AM means they have not yet gone to sleep, i.e. $S > 4$. We want

$$P(\text{DDiet and enough sleep}) = P(\text{DDiet} \mid S > 4)\, P(S > 4,\ W - S \geq 6).$$

The first factor is 0.7 by assumption. For the second factor, consider the joint region $(s, w) \in [1, 6] \times [8, 12]$. We restrict to $s \in [4, 6]$ and $w \geq s + 6$.

For each $s \in [4, 6]$, the allowable w values range from $s + 6$ to 12. Hence the favorable area is

$$\int_4^6 (12 - (s+6))\, ds = \int_4^6 (6-s)\, ds = \left[6s - \tfrac{1}{2}s^2\right]_4^6 = 2.$$

The total area of the rectangle is 20, so

$$P(S > 4,\ W - S \geq 6) = \frac{2}{20} = \frac{1}{10} = 0.1.$$

Therefore

$$P(\text{DDiet and enough sleep}) = 0.7 \times 0.1 = 0.07 = \tfrac{7}{100}.$$

$$\boxed{P(\text{DDiet and enough sleep}) = \tfrac{7}{100}}$$